Vrushali Mhaskar

Software Engineer | Generative AI | LLMs | RAG | Agentic AI Frameworks | Python | AWS | India | 7490044059
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PROFESSIONAL SUMMARY

Strong problem-solving and communication skills with a focus on learning and applying emerging AI technologies. AI-focused Software Engineer with **2+ years of experience** developing **Generative AI applications and data-driven solutions at IBM.** Skilled in Python, Java, React, RAG pipelines, vector databases, semantic search, and LLM integration using LangChain and LangGraph. Experienced in building AI/ML pipelines, integrating AI capabilities with enterprise applications, and developing cloud based solutions on AWS. Focused on delivering reliable, scalable AI solutions with measurable business impact.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, JavaScript, React, Graphite, Streamlit
- **AI Engineering:** Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Agentic AI Frameworks (LangGraph, LangChain), Vector Embeddings, Semantic Retrieval, AI/ML Pipelines, AI Evaluation, Prompt Engineering, LLM Integration, REST APIs,(FastAPI), Data Pipelines, Data Preparation,
- **Databases:** SQL, PostgreSQL, MongoDB, Vector Databases (pgvector)
- **Cloud & DevOps:** AWS Services, Jenkins, CI/CD, Git (version control), Docker
- **Practices:** Software Development Lifecycle (SDLC), Agile Development, Deployment and Monitoring, Cross-Functional Collaboration

PROFESSIONAL EXPERIENCE

Software Developer | *IBM Software Lab, India* | *July 2025 to Present*

- **Develop core modules** of a Python-based **optimization engine** using NumPy and Pandas to generate **10 –11 candidate asset investment plans** per run across long-term planning horizons under user-defined constraints.
- **Improved optimization runtime** by approximately **80%, reducing execution time** from around 5 minutes to 1 minute per scenario on large asset datasets through vectorized NumPy/Pandas operations and efficient data processing.
- **Integrated** the optimization engine with IBM Maximo **through REST APIs**, enabling planners to configure constraints, run scenarios, and evaluate cost-effective maintenance and investment strategies.

Software Developer Intern | *IBM Software Lab, India* | *July 2024 to July 2025*

- Reduced manual QID mapping **effort by 60%** hence reducing **mapping time from 5 days to ~1day** in IBM QRadar, driving process improvement and operational efficiency, by implementing an AI-powered backend that automates security-event categorization.
- Improved **categorization efficiency by 50%** by building Python-based AI/ML pipelines using RAG architecture, vector embeddings, semantic retrieval, and LLM integration to classify unknown security events.
- **Mitigated LLM hallucination** and increased the reliability and safety of generated classifications by implementing retrieval grounded workflows and output safety guardrails that constrain and validate model outputs against security-event context, applying Responsible AI principles.

INNOVATION PROJECTS

AI for Krishi — Agricultural Machinery GenAI Assistant | AI for Bharat

- Designed an end-to-end Generative AI assistant for agricultural machinery, covering document ingestion, data preparation, indexing, semantic retrieval, and LLM-based response generation using AWS Bedrock.
- Built a RAG pipeline integrating document processing, vector embeddings, Amazon OpenSearch, and retrieval-grounded LLM responses to improve access to relevant agricultural knowledge.
- Enabled access in English, Hindi, and Gujarati through AWS services including Transcribe, Lambda, API Gateway, and S3, with CloudWatch monitoring.

EDUCATION

Charusat University | Bachelor of Technology (B.Tech) in Computer Science & Engineering |

CGPA: 9.81/10 | Graduated 2025

CERTIFICATIONS

Microsoft Certified: Azure AI Fundamentals (AI-900)